ARC FLASH HAZARD ANALYSIS PROCESS

IEEE 1584-2018 WORKFLOW

CRITICAL CONSIDERATIONS

- ⚠ MODEL VALID 208V-15KV
- ⚠ THREE-PHASE AC ONLY
- ⚠ MULTIPLE CALCULATIONS NEEDED
- ⚠ SELECT MAXIMUM RESULTS

PHASE 1: DATA COLLECTION

STEP 1: COLLECT SYSTEM & INSTALLATION DATA

- SINGLE-LINE DIAGRAMS
- EQUIPMENT RATINGS
- CONDUCTOR DATA
- PROTECTIVE DEVICE SETTINGS
- INSTRUMENT TRANSFORMER

STEP 2: DETERMINE SYSTEM MODES OF OPERATION

- MAXIMUM FAULT CURRENT
- MINIMUM FAULT CURRENT
- GENERATOR CONFIGURATIONS

STEP 3: CALCULATE BOLTED FAULT CURRENTS

- SHORT-CIRCUIT STUDY
- MULTIPLE SOURCE EFFECTS
- MOTOR CONTRIBUTIONS

PHASE 2: EQUIPMENT ANALYSIS

STEP 4: DETERMINE GAP & ENCLOSURE SIZE

- USE TABLE 8 FOR TYPICAL GAPS
- MEASURE ACTUAL ENCLOSURES
- CLASSIFY ENCLOSURE TYPE

STEP 5: DETERMINE ELECTRODE CONFIGURATION

- VCB: VERTICAL IN BOX
- VCBB: VERTICAL WITH BARRIER
- HCB: HORIZONTAL IN BOX
- VOA: VERTICAL OPEN AIR

STEP 6: DETERMINE WORKING DISTANCES

- USE TABLE 10 FOR TYPICAL
- TASK-SPECIFIC DISTANCES

PHASE 3: CALCULATIONS

STEP 7: CALCULATE ARCING CURRENT

- APPLY EQUATION 1
- CONSIDER VOLTAGE LEVEL

STEP 8: DETERMINE ARC DURATION

- USE TCC CURVES
- FUSE CLEARING TIME
- CB + RELAY TIME

STEP 9: CALCULATE INCIDENT ENERGY

- APPLY EQUATIONS 3-6
- INCLUDE CF CORRECTION
- CALCULATE FOR BOTH IARC VALUES

STEP 10: DETERMINE ARC-FLASH BOUNDARY

- APPLY EQUATIONS 7-10
- 5.0 J/CM² THRESHOLD

PHASE 4: DOCUMENTATION

DOCUMENT RESULTS

- EQUIPMENT LABELS
- INCIDENT ENERGY LEVELS
- ARC-FLASH BOUNDARIES
- PPE REQUIREMENTS

VALIDATE & REVIEW

- CHECK MODEL RANGES
- VERIFY PROTECTIVE DEVICES
- REVIEW ASSUMPTIONS

ANALYSIS COMPLETE